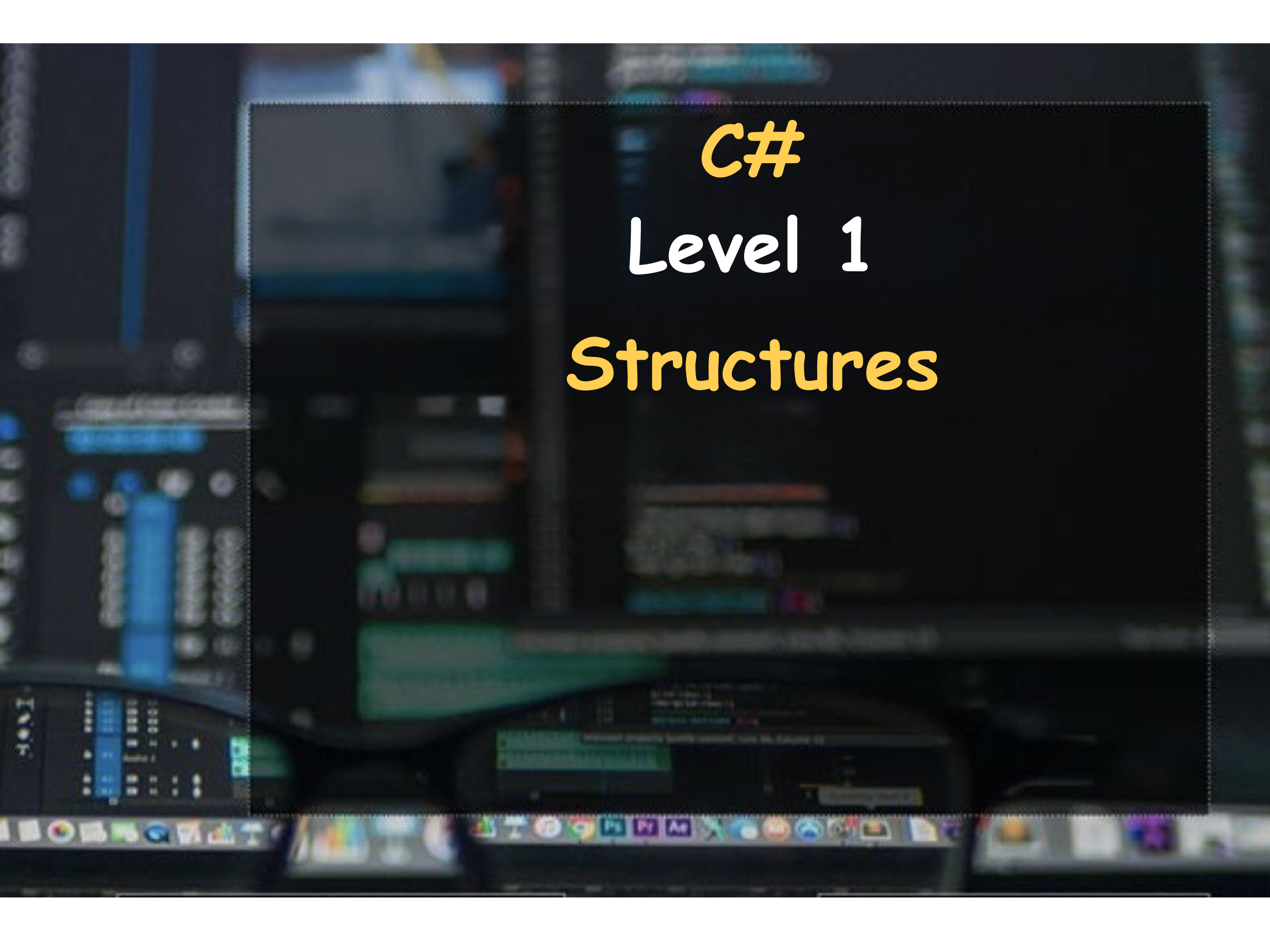


The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, white). At the bottom, a taskbar is visible with several application icons, including what looks like a web browser, a file explorer, and some development tools. The overall image is out of focus, emphasizing the text overlay.

C# Level 1 Structures



C#

Level 1

Structures

Definition

Structure (`struct`)

 A `struct` is a value type that can group related data and functionality into one custom type.

 It is useful for representing small data structures.

Explanation

Defining a Structure

-  A `struct` can contain fields and methods.
-  Fields can be accessed through an object of the struct type.

Procedures

1. Define a `struct`.
2. Add related fields or members.
3. Declare a variable of the struct type.
4. Assign values to its fields.
5. Access the fields using the `.` operator.

💡 Example

```
struct stStudent
{
    public string FirstName;
    public string LastName;
}
stStudent Student;
Student.FirstName = "Dania";
Student.LastName = "Aljaseem";
Console.WriteLine(Student.FirstName);
Console.WriteLine(Student.LastName);

stStudent Student2 = new stStudent();
Student2.FirstName = "Hadi";
Student2.LastName = "Aljaseem";
Console.WriteLine(Student2.FirstName);
Console.WriteLine(Student2.LastName);
```

➡ Result

Dania
Aljaseem
Hadi
Aljaseem

Note

- ◆ ``struct`` is a value type.
- ◆ A struct can contain multiple related values under one type.
- ◆ Struct variables can be created with or without ``new``.
- ◆ When declared without ``new``, all fields must be definitely assigned before they are accessed.

⊘ Warning

- ◆ You cannot read an unassigned field.

Invalid:

```
stStudent Student;
```

```
Console.WriteLine(Student.FirstName);
```

- ◆ The field must be assigned first:

```
Student.FirstName = "Dania";
```



Comparison

Struct

- ✓ Is a value type.
- ✓ Groups related data into a custom type.
- ✓ Can be created with or without `new`.
- ✓ Suitable for small, data-focused types.

Class

- ✓ Is a reference type.
- ✓ Creates objects that are reference-based.
- ✓ Commonly used for larger or more complex objects.

✦ Summary

- ✓ ``struct`` creates a custom value type.
- ✓ It can group related data and functionality.
- ✓ Structs can contain fields and methods.
- ✓ A struct can be created with or without ``new``.
- ✓ Fields must be assigned before they are read.

Thank you 😊